Exercise 2.2. For each of the following, give three examples of this property. Then, prove that it is true.

(a) If n is an even integer, then $-n$ is an even integer.

(b) If n is an odd integer, then $-n$ is an odd integer.

(c) If n is an even integer, then $(-1)^n = 1$. You may use standard properties of exponents.

Solution:

We use the same definitions as before:

- **Even integer:** $n = 2k$, for some $k \in \mathbb{Z}$.
- **Odd integer:** $n = 2k + 1$, for some $k \in \mathbb{Z}$.

(a) If n is an even integer, then $-n$ is an even integer

Three examples

$$n = 2 \quad \Rightarrow \quad -n = -2$$

$$n = 8 \quad \Rightarrow \quad -n = -8$$

$$n = -12 \quad \Rightarrow \quad -n = 12$$

In every case, $-n$ is even.

Proof

Suppose n is even. By definition,

$$n = 2k$$

for some integer k.

Then

$$-n = -(2k)$$

$$-n = 2(-k)$$

Since k is an integer, $-k$ is also an integer.

Therefore, $-n$ has the form

$$2(\text{an integer}),$$

so $-n$ is even.

$$\boxed{\text{If } n \text{ is even, then } -n \text{ is even.}}$$

(b) If n is an odd integer, then $-n$ is an odd integer

Three examples

$$n = 3 \quad \Rightarrow \quad -n = -3$$

$$n = 7 \quad \Rightarrow \quad -n = -7$$

$$n = -5 \quad \Rightarrow \quad -n = 5$$

In every case, $-n$ is odd.

Proof

Suppose n is odd. By definition,

$$n = 2k + 1$$

for some integer k.

Then

$$-n = -(2k + 1)$$

$$-n = -2k - 1$$

We want to write this in the form $2m + 1$. Notice that

$$-2k - 1 = -2k - 2 + 1$$

$$= 2(-k - 1) + 1$$

Since $-k - 1$ is an integer, $-n$ has the form

$$2(\text{an integer}) + 1.$$

Therefore, $-n$ is odd.

$$\boxed{\text{If } n \text{ is odd, then } -n \text{ is odd.}}$$

(c) If n is an even integer, then $(-1)^n = 1$

Three examples

$$(-1)^2 = 1$$
$$(-1)^4 = 1$$
$$(-1)^6 = 1$$

Proof

Suppose n is even. Then, by definition,

$$n = 2k$$

for some integer k.

Therefore,

$$(-1)^n = (-1)^{2k}$$

Using the exponent rule

$$a^{bc} = (a^b)^c,$$

we get

$$(-1)^{2k} = \left((-1)^2\right)^k$$

Since

$$(-1)^2 = 1,$$

we have

$$\left((-1)^2\right)^k = 1^k = 1.$$

Therefore,

$$\boxed{\text{If } n \text{ is even, then } (-1)^n = 1.}$$

The key idea in all three proofs is to **start with the definition of even or odd**, then manipulate the expression until it has the required form.